Experiment - 7

Sliding Window Protocol

AIM:

To implement flow control at the Data Link Layer using Sliding Window Protocol and simulate the transfer of frames between two nodes.

PROCEDURE:

Sender

1. Get the Window size from the user.
2. Read a Text message.
3. Consider one character as one frame.
4. Assign a frame number to each character.
5. Create frames in the format:
   [Frame No., DATA]
6. Store the frames in Sender_Buffer.
7. Send frames according to the window size.
8. Wait for acknowledgement from the receiver.
9. Read Receiver_Buffer.
10. Check the ACK number.
11. If the ACK is correct, transmit the next set of frames.
12. If NACK is received, retransmit the required frame(s).

13. Introduce an error by manually modifying a frame / ACK number and observe the behaviours.

Receiver:

14. Read frames from Sender-Buffer.
15. Check the frame number.
16. If correct, generate ACK
17. If incorrect, generate NACK.
18. Store the response in Receiver-Buffer.

The manual explicitly requires Sender-Buffer, Receiver-Buffer, ACK/NACK handling and induced errors.

Result:

Thus, Sliding Window Protocol was successfully implemented and frame flow control using ACK/NACK was verified.